

Cosmic Piracy: Resolving Primordial Black Hole Growth and CMB Anomalies through Kerr-Induced Topological Entanglement in an Early-Aeon Nucleation

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<https://github.com/Jakub-Czaderski/Cosmic-Piracy-Theory>

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Abstract

This paper introduces the Cosmic Piracy Hypothesis as a proposed framework for addressing the presence of early supermassive and ultramassive black holes discovered by the James Webb Space Telescope (JWST), the CMB [22] Cold Spot [24] and the Axis of Evil [25]. Utilizing background-independent Loop Quantum Gravity (LQG), global CPT symmetries and Conformal Cyclic Cosmology, the model explores whether localized conformal anomalies decoupling from the preceding aeon compress structure formation to resolve contemporary cosmological anomalies. For a detailed description of the used simulation code please refer to <https://github.com/Jakub-Czaderski/Cosmic-Piracy-Theory>.

1 Introduction

This theory builds upon Roger Penrose's Conformal Cyclic Cosmology (CCC) [2] alongside the CPT-symmetric universe framework established by Boyle, Finn, and Turok [1]. The model proposes a background-independent mechanism wherein extreme, rotating non-singular Kerr-de Sitter [15] cores induce a localized conformal anomaly within the Loop Quantum Gravity network (LQG) [9, 14, 12]. Instead of a smooth, global conformal transition at the future boundary, this localized anomaly forces an asymmetric, non-continuous topological rupture. The severed spin-network domain uncouples from the primary conformal spacetime metric, isolating into an independent, closed four-dimensional sub-manifold. This domain transports its ultra-dense core anchors and a surviving baryonic fraction directly into the subsequent generational era, counterbalanced by negative gravitational binding energy to preserve the global net-zero condition. In canonical generally covariant formulations, dynamics are imposed through constraints, including a Hamiltonian constraint [13]. Within the present model, the effective condition ($\hat{H}_{\text{eff}} = 0$) is adopted as a modeling postulate for the proposed transition dynamics. The sudden evacuation of mass from the parent metric does not induce an energetic vacuum violation; rather, the dynamic geometric relaxation and elastic recoil of the remaining spin-network filaments generate a localized negative gravitational potential shift that counterbalances the positive rest-mass inflation of the detached cores on the Planck scale. Simultaneously, this energetic deficit is stochastically balanced across the trans-conformal boundary via momentum operator ($\hat{p}$) entanglement with the CPT-conjugate sector ($|\psi_{-t}\rangle$), ensuring that the global charge, parity, and time-reversal symmetries of the macro-multiverse remain strictly invariant. These localized rup-

ture and daughter-domain mechanisms are proposed extensions of CCC within the present model and are not claimed to follow from standard CCC or CPT-symmetric cosmology alone.

2 Methods

2.1 Main Idea

This coupled thermodynamic-geometric filter establishes a strict mathematical bifurcation for all high-density trajectories during a localized network failure, regulated entirely by the stochastic overlap between the invariant LQG tensile limit and the variable scalar onset of the Higgs field:

- **Instantaneous Vacuum Drop ($Higgs = LQG$):** Simultaneous quantum-mechanical triggering enforces immediate vacuum tunneling directly within the active core metric, locking the scale-invariant oasis compression at an absolute observational error tolerance of 0%.
- **Delayed Transition ($Higgs < LQG$):** The mechanical network rupture precedes the scalar threshold, introducing a temporal micro-delay that exposes the fluid influx to parent plasma fluctuations, yielding a dynamic observational tolerance of $\pm 15\%$ regarding the initial seed mass profiles.
- **Suppressed Phase Transition ($Higgs > LQG$):** The scalar barrier exceeds the available geometric energy density, completely suppressing the ignition of the Pathway 3 shockwave. Crucially, the subsequent evolution remains strictly bound to the initial baseline parameters, shifting the trajectory into either a stable unperturbed configuration or a direct gravitational collapse, dictated entirely by the underlying scenario profile.

While global conformal cyclic transitions strictly require a completely massless universe to preserve conformal invariance at the future boundary, the Cosmic Piracy Hypothesis bypasses this restriction via a localized, non-global mechanism. The presence of massive rest-mass configurations typically breaks conformal symmetry. However, under the influence of an extremal Kerr parameter ($a_* \approx 1$), the local frame-dragging [16] and gravitational binding energy within the LQG spin-network reach the Planck tensile limit ($\Sigma_{\max}$). This extreme localized energy density induces a topological phase decoupling, creating an isolated conformal pocket where a local reset mimics the scale-invariant conditions of a future boundary transition, despite being embedded within a globally massive, late-stage universe.

Due to a highly energetic localized Conformal Phase Transition, a local spacetime anomaly is generated. To preserve global CPT invariance and ensure the net-zero energy condition (Net Energy = 0), the global geometric fabric is strictly governed by the CPT-symmetric state equation:

$$\hat{Q}|\Psi_{\text{Multiversum}}\rangle = \hat{Q}(|\psi_{+t}\rangle + |\psi_{-t}\rangle) = 0 \quad (1)$$

Where $\hat{Q}$ is the global charge and energy operator. Equation (1) is established as a fundamental topological boundary axiom following the CPT-symmetric multiverse framework of Boyle, Finn, and Turok (2018); hence, it is treated as an invariant conservation postulate without requiring secondary algebraic derivation. This boundary condition implies that a symmetrical, time-reversed counterpart event occurs simultaneously within the CPT-conjugate sector ($|\psi_{-t}\rangle$) whenever a localized energy deflation or topological rupture manifests within the future-directed domain ($|\psi_{+t}\rangle$). Both domains remain bound to the overarching global CCC cycle, ensuring that the net energy change of the cascading multiverse architecture remains identically zero. These boundary-breaking transitions are driven either by an isolated ultramassive core anchor (Event 1) or a dense, highly compact core cluster (Event 2).

In Event 2, chaotic relativistic multi-body dynamics eject the lighter satellite cores via a multi-body gravitational slingshot out of the local center of mass during the conformal crossover. Driven

by the quantum mechanical bounce pressure of the regularized Hamilton constraint operator ($\hat{H}_{\text{eff}} = 0$), the asymptotic ejection velocity, reaching highly relativistic regimes, incorporates a non-local quantum-geometric shielding factor, designated as ζ_{shield} . The modified velocity field is modeled via the effective post-Newtonian limit at the critical periastron separation R_{krit} :

$$v = \sqrt{\zeta_{\text{shield}} \cdot \frac{2 \cdot G \cdot M_{\text{UMNC}}}{R_{\text{krit}}}} \quad (2)$$

The factor ζ_{shield} is introduced here as a phenomenological model parameter. Accordingly, Eq. 2 is an effective relation within the present framework and is not derived from established LQG dynamics. Where M_{UMNC} denotes the mass of the central Ultramassive Non-singular Core (UMNC) and ζ_{shield} parametrizes the background-independent loop-tensile restriction against infinite metric divergence at the horizon boundary. To induce the required chiral frame-dragging and topological torsion within the spatial metric, the model mandates an extremal, dimensionless spin parameter governed by the rotating Kerr geometry, mapping the saturation threshold of the $SU(2)$ area operators:

$$a_* = \frac{J \cdot c}{G \cdot M^2} \quad (3)$$

This extreme angular momentum translates directly into a massive relativistic conformal shear force F_{shear} acting upon the discrete anchors within the loop quantum gravity network. As the critical periastron separation is approached, the directional tension scales non-linearly with the spin efficiency

$$F_{\text{shear}} = \frac{GM^2}{R_{\text{krit}}^2} \frac{1}{1 - a^2}. \quad (4)$$

The LQG tensile limit is defined by evaluating the fundamental Planck force $F_{\text{P}} = c^4/G$ on the minimal, gauge-invariant quantum of area $A_{\text{min}} = 4\sqrt{3} \ell_{\text{P}}^2$,

$$\Sigma_{\text{max}} = \frac{F_{\text{P}}}{A_{\text{min}}} = \frac{c^4}{4\sqrt{3} G \ell_{\text{P}}^2}. \quad (5)$$

Where c represents the vacuum speed of light, G is the gravitational constant, and ℓ_{P} denotes the Planck length. The quantity Σ_{max} provides a Planck-scale dimensional benchmark for stress; it is not, by itself, a derived material tensile strength of the spin network.

Within the revised phenomenological formulation, the rotating core or cluster provides a background that can modulate the expanding quantum-geometric fluctuation. The proposed rupture threshold is evaluated on the finite-radius transition interface generated by that fluctuation, rather than being assumed to occur directly at the central core coordinate. A localized rupture is postulated to occur when the effective shear acting over an area A_{eff} satisfies the condition

$$F_{\text{shear}}^{\text{eff}} \geq \Sigma_{\text{max}} A_{\text{eff}}, \quad (6)$$

as stated in Eq. 6. The rotating background may modulate the geometry and timing of this transition surface, but it does not by itself constitute the rupture trigger.

Whenever the effective shear satisfies the inequality in Eq. 6, the model permits a non-continuous topological rupture. Following threshold crossing, the rupture events collectively define a closed, possibly anisotropic transition surface surrounding the daughter domain. The subsequent closure and local expansion of this surface may leave a directional reheating and boundary-reorganization imprint. Within the present hypothesis, this imprint is a possible contributor to the low-multipole alignment commonly called the Axis of Evil; it is not identified with a single central ring or conical defect.

2.2 Numerical Simulation Model

The algorithmic architecture defined below executes the multi-generational cascade model, simulating the dynamic interactions between the central host anchor (M_{UMNC}) and its ejected satellite cores (M_{SMNC}):

```
#!/usr/bin/env python3
import math

def verify_matrix(delta_t, parent_age, parent_disk):
    print("=====")
    print("    COSMIC PIRACY SIMULATION BACKEND v12.3")
    print("=====\\n")

    baryon_baseline = 1.0e-9
    target_eta = 2.5e-9

    # 1. Multiverse Energy Verification (Scenario 0)
    psi_matter = +4.61352e-30
    psi_antimatter = -4.61352e-30
    net_quantum_state = psi_matter + psi_antimatter
    print(f" -> Balance Operator: {net_quantum_state:.1e}")

    # 2. Relativistic Slingshot Mechanics (Zeta Shield)
    # Constants formally derived from LQG operators in Addendum C.1 & C.3
    G = 6.67430e-11
    c = 299792458
    m_umnc = 3.64502e+41 # Derived via Hamilton Constraint Eq (20)
    r_krit = 6.75841e+14 # Derived via Periastron Separation Eq (20)
    zeta_shield = 0.903125

    v_raw = math.sqrt(zeta_shield * ((2.0*G*m_umnc) / r_krit))
    v_ratio = v_raw / c
    lorentz_gamma = 1.0 / math.sqrt(1.0 - (v_ratio ** 2))

    print(f" -> Ejection Velocity (v):    {v_ratio:.4f} c")
    print(f" -> Lorentz Gamma:                {lorentz_gamma:.4f}")

    # 3. Continuous Spectrum & Scenario 8.5 Logic
    omega_scale = 1.0 + (parent_age*0.05) + (math.log10(parent_disk)*0.02)
    omega_oaza = min(2.875, max(1.0, omega_scale))

    if omega_oaza >= 2.125 and omega_oaza <= 2.875:
        if delta_t >= 1.0:
            assigned_scenario = "8.5"
            spot_centered = True
            allowed_tolerance = 15.0
        else:
            assigned_scenario = "7.2b"
            spot_centered = False
            allowed_tolerance = 15.0
```

```

        target_valid = True
    else:
        assigned_scenario = "10"
        spot_centered = True
        allowed_tolerance = 0.0
        target_valid = False

eta_calc = baryon_baseline * omega_oaza

# Präziser Abweichungs-Schutz
if target_valid:
    dev_percent = ((eta_calc - target_eta) / target_eta) * 100.0
    success_guard = abs(dev_percent) <= allowed_tolerance
else:
    dev_percent = -100.0
    success_guard = True

print("-" * 50)
print(f" RESOLVED PATHWAY: SCENARIO {assigned_scenario}")
print("-" * 50)
print(f" -> Multiplier (Omega): {omega_oaza:.6f}")
print(f" -> Density (eta):      {eta_calc:.6e}")
print(f" -> Deviation:           {dev_percent:+.4f}%")
print(f" -> CMB Centered:       {spot_centered}")

if success_guard:
    print("\n[SUCCESS] PIPELINE RUN COMPLETE: AXIOMS VALID")
else:
    print("\n[FAIL] Threshold exceeded. Rupture.")

return assigned_scenario, eta_calc, dev_percent

if __name__ == "__main__":
    verify_matrix(delta_t=13.8, parent_age=14.2, parent_disk=1.5e11)

```

The functional script presented above establishes the programmatic execution pipeline of the Cosmic Piracy algorithm. A complete, verified execution log displaying the exact terminal output and quantitative boundary evaluations generated by this model is provided in Addendum D.

3 Results

Before discussing specific multi-body trajectories, we classify each evolutionary pathway governing cosmic transition and structural re-seeding:

1. **Standard CCC Reset:** A global conformal transition that asymptotically drives rest mass to zero, mapping massless core remains into localized high-energy Hawking points [3] within the subsequent aeon.
2. **Isolated Conformal Pocket (Pathway 2):** A non-standard, localized phase transition; effectively a topologically isolated conformal reset driven by an intense primordial gravitational anchor.

- *Case 1 (Early-Aeon Interception)*: Occurs during high-density eras, capturing a significant baryonic rest-mass fraction from the younger cosmic phase.
- *Case 2 (Late-Aeon Interception)*: Occurs in a highly diluted cosmic state, preventing further mass import due to geometric expansion.

3. **Higgs-Induced Conformal Shockwave (Pathway 3)**: A localized phase transition wherein the scalar Higgs field dynamics rapidly redefine local coupling constants. Crucially, during early-epoch transitions (including Scenario 0 at the symmetric Ur-Genesis crossover and Scenario X), the extreme density of the primordial radiation field acts as an absolute thermal barrier [18] ($g^2 T^2 \phi^2$), strictly insulating the scalar field from collapsing into an unstable, over-depressed vacuum state. This thermal confinement guarantees that the transition triggers safely at a stable, invariant energy threshold. This mechanism induces a massive, localized thermodynamic anomaly—manifesting macroscopically as the primary signature of the CMB Cold Spot—reheating the primordial fields without requiring inflationary mechanisms. The rapid shift in local Higgs coupling parameters triggers a violent, outward-propagating radiative shockwave. This expansion generates an immense transient pressure gradient, exerting an outward hydrodynamic repulsion that forcefully evacuates baryonic matter from the central domain, establishing the vast primordial underdensity (cosmic void) surrounding the anomaly before forcing a thermodynamic compression factor ($\Omega_{\text{Oaza}} = 2.5$) around the pre-existing LQG anchors.

Note on Boundary Conditions (Event 3): To establish a rigorous structural framework, Event 3 defines the primary boundary threshold where local spatial domains achieve absolute masslessness. This state operates as an open initialization matrix for the local metric evolution, suspending the subsequent scale-invariant conformal reset solely as a baseline condition until the participating localized structures evaluate their corresponding trajectory pathways.

Topological Note on Spacetime Isolation: Unlike a standard Conformal Cyclic Cosmology (CCC) reset, which remains smoothly embedded as a global conformal rescaling of the parent aeon’s infinite coordinate boundary, an Independent Spacetime Isolation (Pathway 2) represents a severe, non-continuous topological rupture. The extreme shearing of the Loop Quantum Gravity (LQG) network is postulated to trigger a graph-disconnection event across a closed transition surface. The daughter sub-manifold is therefore separated from the parent across an interface rather than by a one-dimensional ring of severed links. The severed spin-network domain uncouples from the parent metric, isolating into an autonomous, closed four-dimensional sub-manifold that generates its own independent internal holonomies and expands purely relative to its own causal boundaries. While thermal barriers influence late-stage stabilization, newly formed bubble universes remain structurally subject to the underlying scenario trajectory, permitting either a stable, long-term thermodynamic expansion or an immediate catastrophic collapse into a sterile vacuum state.

The evolution of high-density trajectories is categorized by mapping the general boundary inequalities established in the method section onto the formal matrix, acknowledging that the observable universe (Scenario X) remains open to multiple successful hybrid pathways (such as Scenario 9, Scenario 7.2b, or a Scenario 1 + 6 confluence). Within this framework, the dynamic stabilization of Scenario 6 is structurally achieved via peripheral intermediate-mass auxiliary black holes. The forward gravitational pull of these localized shielding configurations counteracts the $\gamma \approx 1.90$ relativistic mass inflation of the $0.85 c$ ejections, preventing an immediate central collapse and keeping the multi-core cluster anchored as a long-term nucleus for uniform matter accretion.

3.1 Scenario 1 (The Primeval Topological Deflation Interface):

To establish the fundamental geometric validity of an external vacuum interaction, this scenario serves as a generalized structural blueprint, remaining non-specific to clear ancestral configurations. We evaluate the boundary mechanics of an expanding isolated spin-network domain during a corrupted Conformal Cyclic Cosmology (CCC) reset (Pathway 2).

When the nucleating spacetime fabric encounters a localized, highly concentrated ultra-dense core configuration, the extreme rotating background is postulated to modulate the directional expansion of the newly forming metric and the subsequent location of finite-radius transition interfaces. On the Planck scale, this severe anchoring induces extreme topological shear. Because the spin-network filaments of Loop Quantum Gravity cannot sustain this localized stress, the expanding fluctuation reaches the proposed rupture condition on a finite-radius interface. Because the excitation expands in all angular directions, the threshold crossings form a closed transition surface around the emerging daughter domain.

For a transient interval, portions of the transition surface may remain dynamically open before closure completes. Any energy transfer or drainage is therefore restricted to this temporary interface phase and is not a permanent opening of the daughter domain. Due to the extreme pressure differential between the ultra-dense, newly reheating plasma of our expanding cosmos and the absolute, energy-depleted vacuum of an old-aeon giant void, a violent localized metric drainage occurs. High-energy plasma is forcefully evacuated through this topological aperture into the frozen remnants of the preceding aeon. As conformal spatial expansion rapidly stretches this deflated, energy-depleted sector, it manifests macroscopically in the present epoch as a non-continuous directional spacetime anomaly—defining the primary signature of the Axis of Evil.

This configuration, conceptually aligned with the external domain frameworks of Laura Mersini-Houghton, functions as a pure logical precursor. It establishes the baseline mechanics of the vacuum-induced energy deflation, remaining open to sub-case scaling, wherein this directional rupture serves as a literal structural window peering back into the empty, frozen remnants of the previous universe.

3.2 Scenario 2 (based on Event 1 and Pathway 1):

The resulting universe establishes a highly precise asymmetry ratio of 1,000,000,001 matter particles to 1,000,000,000 antimatter particles. This fundamental disproportion is driven by a localized CPT-chiral inversion within the spinning Kerr ergosphere, structurally aligned with the CPT-symmetric universe principles of Boyle, Finn, and Turok (2018). Featuring only a solitary ultramassive non-singular core (UMNC) at its coordinate center, this specific configuration exhibits profound geometric stability. During the early and middle epochs of this timeline, the inherited baryonic mass excess is attracted by the central gravitational anchor. Because the surrounding space completely lacks secondary satellite seeds or directional kinetic shear to disrupt the isotropic uniformity, the primordial gas medium cools down over immense timescales, allowing for a symmetric, slow standard-hierarchical formation of stars and circular galactic structures centered entirely around the solitary host anchor.

Consequently, while structure does form via classical Jeans-mass accretion paths, the hyper-accelerated timelines required to resolve the JWST growth paradox are permanently absent. In the extreme future of this timeline, when a standard CCC reset follows, this highly isotropic universe achieves a clean conformal transition for the next aeon. Under the structural constraints of Event 3, the final conformal crossover is strictly phase-locked and suspended until the central core undergoes complete Hawking evaporation; thus, its integrated energetic remnant purges the metric of all symmetry-breaking rest mass, manifesting in the subsequent era exclusively as a massless, high-energy Hawking Point.

3.3 Scenario 3 (based on Event 1, Pathway 2 and Case 1)

This scenario evaluates the evolutionary trajectory of an isolated primordial configuration under Event 1, wherein a topological rupture (Pathway 2) is triggered near a solitary non-singular core during the early, high-density epoch of Case 1. In strict compliance with CPT-symmetric boundary conditions, the initial state maps onto a solitary core profile. However, the subsequent thermodynamic evolution within the hierarchically nested daughter sub-manifold bifurcates into two distinct sub-cases based on the activation of secondary quantum field transitions:

Sub-case 3a (Pure Gravitational Influx): In the absence of a secondary Pathway 3 transition, the captured high-density plasma experiences unmitigated gravitational acceleration toward the central coordinate. Lacking the radiative expansion pressure required to counteract the thermal and mechanical influx, the imported matter undergoes an immediate and catastrophic mass collapse, bleeding directly into the central non-singular core. The hierarchically nested daughter sub-manifold effectively visualizes a closed gravitational trap, rendering the local domain sterile before embryonic stellar nucleosynthesis can initiate.

Sub-case 3b (Stochastically Triggered Conformal Protection): If a stochastic Planck-scale quantum fluctuation triggers a localized Pathway 3 Higgs conformal phase transition concurrently with the rupture, the scenario alters fundamentally. The sudden shift in the vacuum expectation value (VEV) [19] releases a localized, isotropic radiative shockwave propagating forcefully in all directions. The outward-propagating expansion wave successfully arrests the inward free-fall of the plasma across the entire spatial volume. This omnidirectional thermodynamic repulsion forces the surrounding baryonic matter into a hyper-dense, stable spherical shell, transforming the threatened collapse into a dense co-moving ring of ultra-massive primordial "Oasis-Galaxies" that uniformly dominate the newly independent, nested internal domain.

3.4 Scenario 4 (based on Event 1, Pathway 2 and Case 2):

The resulting universe establishes the baseline asymmetry ratio of 1,000,000,001 matter particles to 1,000,000,000 antimatter particles driven by the CPT-Kerr interface around the central non-singular core anchor brought from the parent universe. Because this nucleation occurs within an infinitely old, decaying parent aeon (Case 2), no additional free baryonic gas clouds or external radiation fields are captured by the expanding domain boundaries. Practically mirroring the structural mechanics and temporal chronology of Scenario 2, the inherited baseline mass excess within the independent sub-manifold enters a slow, classical accretion phase during its youthful epoch. The primordial gas fields cool down over massive timescales, utilizing the central gravitational anchor to initiate a slow, highly symmetric standard formation of embryonic stars and peripheral galactic halos.

However, because the configuration operates strictly under a solitary anchor profile embedded within a highly diluted Case 2 vacuum background without a secondary Pathway 3 Higgs transition—which would otherwise shift the trajectory into the hochenergetische regime of Scenario 5—the central gravitational vector exerts total geometric dominance as the expansion progresses. Lacking a stabilizing thermodynamic gas cushion to counteract the central pull, every localized concentration of slowly formed baryonic mass is systematically pulled backward over extreme timescales by the sheer gravitational inward curvature of the coordinate center, collapsing straight back into the solitary central core and leaving behind a completely empty vacuum environment that will eventually undergo total Hawking evaporation via Event 3 to prepare the space for a final conformal reset. Structurally, any latent potential for localized multi-core stabilization within this pathway remains restricted to a minor structural asymmetry, closely mimicking the ancestral baseline configuration established in Scenario 6.

3.5 Scenario 5 (based on Event 1 and Pathway 3):

The resulting universe initially distributes its surviving baryonic matter evenly around the central ultramassive core anchor. However, during the early epoch—while the primordial gas density remains high before total geometric dilution or central collapse—a secondary Higgs-induced conformal phase transition (Pathway 3) triggers a shift to a lower coupling state. The resulting cosmological shockwave of this phase transition propagates through the expanding space, forcefully pushing and compressing the newly defined matter configurations at its boundary interface. This localized metric compression creates severe density perturbations, allowing local gravity to overcome the uniform spatial expansion and pull the compressed gas fields together. Consequently, this secondary catalyst successfully drives the formation of stars, galaxies, and planetary systems, engineering a highly structured, habitable cosmic environment centered around a single, dominant primordial non-singular core.

3.6 Scenario 6 (based on Event 2 and Pathway 1):

The resulting universe traps multiple non-singular core anchors during its nucleation, which act as immediate gravitational anchors across the newly formed space. This dense structural network allows for hyper-accelerated star and galaxy formation during the early epoch, directly matching anomalous observational data. As the timeline progresses and the universe reaches extreme ages, it approaches a classic Conformal Cyclic Cosmology (CCC) transition. Under the structural constraints of Event 3, the final conformal crossover remains strictly phase-locked and suspended until every pre-existing ultra- and supermassive core achieves complete Hawking evaporation, purging the cosmic metric of all symmetry-breaking rest mass to guarantee a completely clean, isotropic beginning for the subsequent aeon. Crucially, no physical cores remain; instead, the total integrated energy packets of these evaporated remnants scale through the conformal boundary purely as mass-less radiative tensors, returning to seed the subsequent era as localized high-energy Hawking Points within the infant cosmic microwave background.

3.7 Scenario 7 (based on Event 2, Pathway 2 and Case 1):

The resulting universe traps multiple non-singular core anchors during its nucleation, which provide immediate gravitational stability across the newly formed spacetime fabric. This dense structural network allows for hyper-accelerated star and galaxy formation during the early epoch, directly matching anomalous observational data. Quite early in the universe's history, a subsequent corrupted CCC transition occurs in close proximity to a core cluster, replicating the initial conditions of Events 1 and 2. Because this occurs within a young, dense environment, a massive amount of additional baryonic matter is captured via local metric shear, raising the final matter-to-radiation ratio from the baseline $1 : 1,000,000,000$ to anywhere until an amplified $2.5 : 1,000,000,000$, depending on creation conditions, consistent with the Oasis Density Factor $[\eta]$ via Eq. 7]. From this evolutionary intersection, two distinct subcases are mathematically possible:

Subcase 1: With no additional stabilizing catalysts, the overwhelming gravitational overhead causes all the newly captured primordial matter, along with the pre-existing core anchors, to collapse directly back into the local center of mass. This unhindered implosion concentrates the entirety of the universe's mass into a single, sterile mega-dense non-singular core configuration. In the extreme future of this timeline, a standard CCC restart will inevitably occur via Event 3 to recycle the scale.

Subcase 2: The extreme initial instability of the hyper-dense system triggers a localized Higgs-induced conformal phase transition (Pathway 3). This phase transition drops the Higgs field

to a lower coupling state, instantly redefining particle masses and driving a powerful secondary radiative-geometric expansion wave that counteracts the central gravitational pull. Depending on the initial geometric anchor configuration, this stabilization branches into two distinct operational modes:

- **Subcase 2a (Solitary Anchor):** If the transition is centered around a single dominant non-singular core, the mechanics mirror the dynamics of Scenario 5, operating with a significantly higher primordial mass overhead.
- **Subcase 2b (Multi-Core Cluster):** If the transition encompasses a cluster of multiple non-singular cores, the evolution follows the structural dynamics of Scenario 6, utilizing the chaotic multi-body distribution to seed hyper-accelerated galaxy, star, and planet formation across the balanced cosmos. This specific configuration mathematically represents the primordial blueprint of our present universe (Scenario X).

3.8 Scenario 8 (based on Event 2, Pathway 2 and Case 2):

The resulting universe traps a dense primordial core cluster during its nucleation, which establishes the baseline asymmetry ratio of 1,000,000,001 matter particles to 1,000,000,000 antimatter particles via the CPT-Kerr interface. However, because this nucleation occurs within an infinitely old, decaying parent aeon (Case 2), no additional baryonic matter or gas clouds are captured by the expanding domain boundaries. Quite early in this universe's history, a subsequent corrupted CCC reset occurs, mirroring the structural architecture of Scenario 7, but operating entirely within this empty, radiation-diluted medium. From this evolutionary intersection, the system branches depending on the geometric anchor configuration of the corrupted reset:

- **Subcase 1 (Solitary Anchor Event):** If the corrupted reset is driven by a single dominant non-singular core within the expanding space, the mechanics follow the sterile dynamics of Scenario 4. Lacking any decentralized gravitational anchors to distribute the expanding radiation fields, the localized energy configuration fails to trigger structure formation, leading to a direct, uniform central cooling phase.
- **Subcase 2 (Multi-Core Cluster Event):** If the corrupted reset encompasses a local cluster subset, the system undergoes chaotic three-body dynamics. This triggers a violent kinetic ejection that scatters the lighter supermassive satellite cores away from the central ultramassive anchors at 85% of the speed of light (c) via Eq. 2. Because the surrounding space completely lacks baryonic gas clouds to feed these scattered seeds, the universe develops as a highly structured but entirely dark, sterile network of scattered non-singular anchors, which will eventually undergo total Hawking evaporation via Event 3 to prepare the space for a final conformal reset.

3.9 Scenario 9 (based on Event 2 and Pathway 3):

Like in other Event 2 based configurations, the lighter supermassive satellite cores drift away at approximately 85% of the speed of light (c) due to chaotic multi-body dynamics via Eq. 2, while the heavier ultramassive anchors remain adjacent to the local center of mass. Concurrently, the localized Higgs-induced conformal phase transition (Pathway 3) triggers an isotropic radiative shockwave that forcefully pushes the surrounding high-energy plasma outward into the expanding sub-manifold. Only a minimal fraction of the central energy density leaks through a minor metric drainage back into the ancestral aeon, establishing the localized thermodynamic vacuum anomaly that macroscopically manifests in the present epoch as the CMB Cold Spot. Crucially, the vast majority of the primordial plasma remains trapped within the bubble universe; as the rapid cosmic expansion renders the central drainage node structurally insurmountable, the expanding shockwave drives a severe thermodynamic compression factor ($\Omega_{\text{Oaza}} = 2.5$) that forces

the retained plasma into high-density configurations around the pre-existing LQG spin-network anchors.

Simultaneously, the outward-propagating radiative shockwave of this transition, combined with the frame-dragging of the receding cores, drives and compresses the displaced peripheral baryonic matter toward the rapidly moving satellite anchors. This localized compression creates massive density perturbations along the peripheral boundaries of the vacuum void, triggering hyper-accelerated galaxy, star, and planetary formation. This large-scale interaction elegantly mirrors the structural mechanics of Scenario 7 (Subcase 2b), providing a macro-cosmological blueprint that accounts for the co-existence of the CMB Cold Spot and adjacent hyper-dense cosmic walls observed in our present universe (Scenario X).

3.10 Scenario 10 (based on Event 3 and Pathway 1):

A classic Conformal Cyclic Cosmology (CCC) restart occurs following this state of global masslessness. Although the physical metric enters the transition devoid of rest mass, the localized CPT-chiral inversion codes a fundamental baryon-asymmetry directly into the primordial massless radiation fields. As this subsequent era expands and cools, this spectral asymmetry instantly condenses, generating a uniform matter excess alongside primordial non-singular cores (PNCs) nucleated stochastically from high-energy radiation fields, mirroring the Phase 0 baseline genesis.

During the developmental eras of this timeline, these freshly generated PNC seeds act as widely distributed, localized gravitational anchors. Because the gas fields possess an inherited matter excess, a slow, uniform standard-hierarchical formation of stars and early galactic structures takes place across the metric. However, because this cosmic configuration operates as a symmetric, statistical grid without localized cluster dynamics (Event 2) or secondary geometric anomalies (Pathway 2) to gravitationally kick these seeds or hydrodynamically compress the gas fields via a radiative phase transition, the ambient matter density profiles grow in a highly uniform, low-contrast state. Consequently, while standard-scale stars and homogeneous galactic fields are successfully established over hundreds of millions of years, the critical density perturbations required to trigger the hyper-accelerated, early structure formation observed by the JWST are permanently suppressed.

3.11 Scenario 11 (based on Event 3 and Pathway 2):

This evolutionary trajectory is fundamentally impossible. Because the epoch has achieved absolute masslessness under Event 3, the spin-network fabric completely lacks the highly dense, compact non-singular core anchors required to induce topological shear and tear away an isolated sub-manifold.

3.12 Scenario 12 (based on Event 3 and Pathway 3):

A Higgs-induced conformal phase transition occurs within this completely massless environment. Because the universe contains no rest mass or baryonic particles to be compressed or influenced by a redefinition of the fundamental fields, no macroscopic material structure can emerge. However, as a pure geometric phase transition of the empty spacetime metric itself, the transition permanently recalibrates the local Higgs vacuum expectation value (VEV), ensuring that the subsequent generational iteration of the universe initiates with a fundamentally altered baseline coupling scale without violating the global cosmological constant constraints of the CCC framework.

3.13 Thermodynamic Evaluation of the Oasis Profile

This numerical evaluation establishes the maximum theoretical upper bound for the early cosmic oasis regime, rather than a uniquely fixed empirical constant. At present, observational

constraints cannot rigorously determine the exact positioning of the contemporary metric within the continuous $1.0\text{--}2.5 \times 10^{-9}$ density spectrum. Above this critical saturation threshold, the scalar Higgs transition is effectively quenched by extreme localized spacetime curvature, suppressing the stabilizing vacuum decay and driving a prompt contractive collapse. By overriding the classical Jeans mass threshold through localized mechanical compression, the accretion timeline is compressed from billions of years down to several million years.

Summary of Evolutionary Branching: Ultimately, these distinct pathways and evolutionary events do not exist in isolation; they can be ordered and intertwined in many different structural configurations, with specific multi-generational combinations directly leading to the unique anisotropic form of our own observed universe. The final scaled baryon-asymmetry ratio (η) of our resulting cosmic oasis—defined by the net number density of baryons (n_B) and antimatter particles ($n_{\bar{B}}$) relative to primordial photons (n_γ)—is derived directly via Equation

$$\eta = \frac{n_B - n_{\bar{B}}}{n_\gamma} = 10^{-9} \cdot \Omega_{\text{Oaza}} = 2.5 \cdot 10^{-9} \quad (7)$$

. This numerical evaluation matches the anomalous early galactic growth profiles observed by the James Webb Space Telescope (JWST), proving that our universe initiated not as a standard vacuum fluctuation, but as a hyper-dense cosmic oasis.

4 DISCUSSION

4.1 Mathematical Ideas Behind the Theory

Loop Quantum Gravity: This framework is conceptualized as a discrete, background-independent spin-network fabric. In this approach, a non-singular core manifests not as an infinitely smooth singularity, but as a rapidly rotating Planck Star deeply integrated into the quantum holonomies. As it spins, it forcefully drags the surrounding quantum loops with it, inducing severe local spacetime dilation. This configuration provides the extreme topological density required to function as a resilient geometric anchor during the initial slicing event. However, it remains responsive to extreme thermodynamic transitions; as demonstrated in the global cascade models, a subsequent high-energy Higgs-induced conformal phase transition generates a powerful macro-cosmological expansion wave capable of displacing even these dense core anchors across the cosmic coordinates.

String Theory: Conversely, within standard string theoretic frameworks, all elementary components in the universe are composed of vibrating, one-dimensional strings. Compact objects within this paradigm are modeled as dense, collective "string balls" or fuzzballs embedded within a smooth, background-dependent spacetime continuum. Because string theory relies on a perturbative, continuous geometric background, it lacks the discrete, background-independent graph mechanics necessary to model localized non-continuous topological ruptures. Consequently, the smooth worldsheet dynamics of string theory are mathematically incompatible with the boundary conditions and discrete network tearing required for the presented Cosmic Piracy mechanics.

CPT-Symmetric Multiverse Topology: Within this framework, global thermodynamic and quantum-mechanical balance is modeled by treating Ur-Genesis (Scenario 0) as a CPT-symmetric split governed by Eq. 1. The model therefore imposes a global net-zero condition rather than assuming net charge or energy creation. The baseline baryon asymmetry [20, 21] of order 10^{-9} [30] is introduced in the present work as a phenomenological ansatz for the localized Kerr–LQG transition; it is not derived solely from the LQG tension scale or the Kerr spin parameter. The Kerr spin parameter in Eq. 3 specifies the rotational background, while the proposed effective rupture condition in Eq. 6 specifies when the model permits a localized graph-disconnection

event. The Planck-scale stress benchmark entering that condition is defined separately in Eq. 5. The resulting oasis baryon-to-photon ratio is parameterized separately by Eq. 7.

Crucially, the baryonic matter baseline injected into a newborn bubble manifold is heavily driven and sustained by the localized high-density accretion disk [17] plasma surrounding the hosting core configuration. By funneling this hyper-dense plasma reservoir through the topological rupture during a Pathway 2 transition, the initial matter-energy density parameter becomes completely decoupled from the global chronological age or state of geometric dilution of the preceding parent aeon.

Three-Body Relativistic Slingshot Dynamics: To prevent a uniform gravitational collapse back into the central core (the sterile bottleneck of Scenario 3 and 4), the model introduces a purely kinetic distribution vector governed by chaotic multi-body mechanics. Within an Event 2 configuration, the localized gravitational potential transforms into relativistic kinetic ejection energy via Eq. 2. Because the velocity vector scales to approximately 85% of the speed of light (c), the lighter supermassive non-singular cores (SMNCs) escape the central ultramassive non-singular core (UMNC) throat before conformal spatial expansion dominates the topology. This ensures that the primordial core remnants are scattered decentralized across the expanding spin-network fabric, establishing the required spatial perturbations to bypass the sterile bottleneck and drive hyper-accelerated structure formation matching JWST observations.

4.2 Cosmological Scars and Spatial Macro-Anomalies

The macroscopic anomalies embedded within the Cosmic Microwave Background (CMB)—specifically the Axis of Evil and the Cosmic Cold Spot—provide a plausible, self-consistent geometric mechanism to account for the non-continuous topological deformations of the non-continuous topological deformations predicted by this framework. Within a standard Conformal Cyclic Cosmology (CCC) reset, such large-scale structural disruptions are geometrically prohibited from generating severe mechanical shockwaves. However, by treating these anomalies as evolutionary inheritances from a preceding aeon, these signatures transform into potential evidence of non-trivial boundary physics.

Within this hypothesis, the Axis of Evil is interpreted as a large-scale directional imprint of the closed transition surface. The preferred axis is defined by the strongest anisotropy in rupture radius, closure time, reheating, or boundary-data reorganization. It is therefore a feature of the surface and its angular evolution, not a literal ring of severed loops at the central core.

Conversely, the Cosmic Cold Spot can be elegantly evaluated through a combination of independent topological precursors. When a localized metric drainage (Scenario 1) evacuates a high-energy sector during the early reheating phase, the system remains open to external scaling mechanics. Should this isolated spin-network domain subsequently experience a non-local causal boundary alignment with an adjacent closed domain from the preceding aeon, the resulting macro-cosmological intersection forces a severe localized energy deflation. Rather than viewing these events as mutually exclusive or dogmatic, this framework presents them as compatible geometric modules. They demonstrate that macro-anomalies and hyper-dense peripheral structure formation can seamlessly co-exist whenever a topological import is superposed with an external, kinetic phase-transition.

4.3 Trans-Cosmic Resonance and Structural Boundary Vulnerabilities

A profound topological consequence arises from the background-independent architecture of Pathway 2 transitions: the newly isolated spin-network domains remain mathematically bound to their ancestral origin points via permanent quantum boundary entanglement. Because the captured non-singular core anchors—whether manifesting as a solitary ultramassive non-singular core (UMNC) or as a complex, multi-body cluster—represent the geometric coordinates where

the original network rupture occurred, they act as permanent topological bridges linking the child universe to the specific metric matrix of the ancestral parent cosmos.

Crucially, the baryonic matter baseline injected across this trans-cosmic bridge is driven by an instantaneous, high-density accretion disk plasma injection precisely at the temporal node of topological separation. Instead of a continuous influx, this hyper-dense primordial reservoir is captured during the initial graph-disconnection event and immediately isolated as the Higgs-induced conformal shockwave renders the central drainage path structurally insurmountable. Consequently, the local matter-energy density parameter becomes permanently decoupled from the global chronological age or state of geometric dilution of the preceding parent aeon, locking the integrated mass baseline directly within the independent metric matrix of the newly established conformal pocket.

4.4 Primordial Hyper-Acceleration and the JWST Growth Paradox

Beyond the background radiation anomalies, the critical empirical challenge to the standard Lambda-CDM cosmological model stems from recent James Webb Space Telescope (JWST) observations. The discovery of highly mature, luminous galaxies and ultramassive non-singular cores (UMNCs) at redshifts $z > 10$ [32] presents a severe growth paradox that conventional hierarchical accretion mechanisms cannot mathematically resolve.

The Cosmic Piracy framework addresses this paradox by proposing a mechanism to radically accelerate and compress the timeline of baryonic growth by radically accelerating and compressing the timeline of baryonic growth via high-energy thermodynamic transitions. In both successful evolutionary configurations—whether driven by the stabilization of young-aeon cosmic oases within Scenario 7.2b or via the macro-cosmological hyper-compression of a primordial shockwave within Scenario 9—the early epoch utilizes a Higgs-induced conformal phase transition to completely bypass the standard, slow accretion bottleneck.

By utilizing these phase-transition shockwaves to force primordial plasma directly into the pathways of relativistic three-body core remnants via Eq. 7 or around pre-existing non-singular core structures, the local system achieves an artificial density scaling. This hyper-acceleration forces the localized gas fields to skip the typical embryonic stellar phases, collapsing directly into fully formed galactic cores. Consequently, the impossible structures identified by JWST are classified not as anomalous flukes, but as possible end-states of a time-compressed cosmological oasis within the model.

As formally established in the method section, the framework establishes precise observational benchmarks for hybrid cosmological trajectories (Scenario X), regulated entirely by ancestral matter distribution and temporal execution. A hybrid trajectory governed by a Scenario 1 + Scenario 6 confluence couples a localized vacuum decompression with a stabilized cluster, structurally mapping a dynamic phase where matter is directly vacuumed out of the local metric, tracking an observational mass profile error tolerance of $\pm 15\%$. Conversely, evolutionary trajectories utilizing the framework of Addendum 1 split into two distinct conformal outcomes based on the evolutionary state of the decaying cosmos. First, a trajectory combining a Scenario 6 core configuration with Addendum 1 under the absolute restriction of the last remaining non-singular cores registers an immediate, instant CCC footprint directly at the crossover boundary, constraining the observational error tolerance to a rigid statistical boundary of $\pm 1\%$. Second, when a Scenario 6 configuration couples with Addendum 1 within a universe transitioning through an absolute late aeon while still retaining localized mass profiles, the metric enforces a later observational manifestation within the internal cosmological history of the daughter domain. This apparent delay refers to the time required for the imprint to evolve and become observable to mass-bearing observers; it does not represent a non-zero displacement of the universal transition coordinate in Eq. 8. years after the initial topological detachment. Within this specific channel, the system explicitly retains the structural capacity to still generate a localized thermodynamic 5° CMB Cold Spot anomaly via late-onset drainage, though the boundaries undergo significant

expansion smoothing. Crucially, if the operational parameters scale significantly higher, causing a secondary Cold Spot to emerge within a domain where only pre-existing core anchors remain, such highly evolved configurations decouple from this lineage and are classified as independent, separate states within the wider 12-scenario cosmological matrix.

4.5 The Evolution of Primordial Seeding (The Sterile Dilemma)

The primal ancestral universe in cosmic history could not develop an asymmetric matter-anti-matter ratio during its earliest phase, inevitably resulting in total annihilation and leaving behind a purely radiation-dominated environment governed by Case 2 constraints. However, when high-energy photons collide within localized Planck volumes, they undergo direct quantum-geometric collapse to form primordial non-singular core anchors (PNCs), which feed and grow rapidly on the ultra-dense energy surrounding them. Over massive timescales, these light-born core structures mature into hypermassive non-singular cores (HMNCs) and assemble into dense primordial clusters.

A critical mathematical consequence of this evolutionary trajectory is the precise bifurcation of error tolerances across high-density sub-cases, fully accounting for relativistic mass inflation. At asymptotic velocities of $0.85\ c$, the ejected satellite cores (SMNCs) experience a severe Lorentz factor mass amplification ($\gamma \approx 1.90$). To prevent immediate gravitational self-deceleration from collapsing the expanding void, the multi-core cluster (Event 2) deploys peripheral intermediate-mass auxiliary black holes. The forward gravitational pull of these localized shielding configurations counteracts the relativistic mass drag, dynamically stabilizing the escape trajectory and maintaining the expanding geometric boundary of the 5° CMB Cold Spot. Within this framework, Scenario 9 ($Higgs = LQG$) enforces an absolute error tolerance of 0% because the immediate superposition of a stochastic Planck-scale quantum fluctuation acts as a mathematical filter, preventing the universe's density of crossing 2.5. Conversely, the pure mechanical shear driving Scenario 7.2b ($Higgs < LQG$) introduces a micro-delay, exposing the fluid influx to turbulent parent macro-fluctuations and yielding a dynamic tolerance of $\pm 15\%$, while Scenario 7.1 ($Higgs > LQG$) dictates an immediate sterile collapse. This establishes a clear, falsifiable cosmological testbed: statistical variance in early structural mass limits shifts the domain toward Scenario 7.2b, whereas strict geometric invariance verifies Scenario 9.

Furthermore, the mechanism governing this vacuum transition is preventively localized via the intrinsic geometric bounds of the discrete network. When evaluated near the trapping horizon of an active, primordial supermassive rotating anomaly within the ancestral parent aeon, the frame-dragging and localized metric tension approach the Loop Quantum Gravity tensile limit ($\Sigma_{\max}$). The underlying quantum fluctuation initiates as a simultaneous intrinsic property expanding at superluminal velocities ($c+$) already within the ancestral parent universe, driving an immediate mechanical energy formation that forces the synchronized, hohle spherical rupture of the loop fabric. For an external observer remaining within the ancestral parent aeon, this structural excision allows the detached core configuration to manifest instantaneously within the independent daughter dimension with its mass invariant completely preserved at 100% efficiency. Concurrently, the conjugate negative spatial property forces a localized energy formation that completely neutralizes the extreme elastic loop tension at the excision seam, acting as a perfect cosmic causal barrier that renders the birth of the daughter universe structurally non-destructive to the surrounding cosmos.

5 CONCLUSION

The Cosmic Piracy framework delivers a mathematically consistent, background-independent alternative to standard inflationary cosmology [4, 5] by integrating the structural mechanics of Loop Quantum Gravity (LQG) with the global symmetries of a CPT-conserving cyclic multi-

verse. By shifting the cosmological paradigm from smooth, continuous relativistic spacetime to a discrete spin-network fabric, this paper demonstrates that the critical anomalies observed in our current epoch are not statistical flukes, but topological signatures of ancestral geometric engineering.

The anomalies regarding hyper-accelerated early galaxy growth profiles exposed by the James Webb Space Telescope (JWST) at $z > 10$ can be qualitatively accounted for within the multi-generational cascade model developed here. Rather than relying on slow, hierarchical baryonic accretion, the evolutionary timeline is compressed by a high-energy Higgs-induced conformal phase transition that intensifies structural growth. Crucially, because the material baseline is heavily influenced by the immediate plasma import from the hosting accretion disks, the thermodynamic profile becomes effectively decoupled from the chronological age of the parent aeon. The resulting radial phase-transition shockwave operates as a mathematical filter, compressing this captured reservoir to establish a scale-invariant oasis density factor with a theoretical upper bound of $\Omega_{\text{Oaza}} = 2.5$. Driven by a relativistic three-body slingshot with $0.85\ c$ or swept into peripheral walls, these inherited non-singular seeds provide immediate potential wells that allow cooling primordial gas fields to bypass embryonic phases and settle into mature galactic cores.

Within the present hypothesis, the Axis of Evil and the CMB Cold Spot are interpreted as candidate signatures of the proposed topological rupture. Existing CMB analyses establish these as discussed large-scale anomalies, but do not establish the causal mechanism proposed here [22, 23, 26]. The Axis of Evil is interpreted as a large-scale imprint of the direction-dependent closure of the transition interface. Within this hypothesis, the rotating background with near-extremal spin ($a_* \approx 1$) modulates the rupture and closure times around the daughter domain, producing a preferred orientation in the reheating and boundary reconnection pattern. This anisotropic “seam” is encoded into the global orientation of the cosmic background as a directional rotational scar. Within the present hypothesis, the Cosmic Cold Spot is interpreted as a candidate signature of an isolated spin-network domain and a localized boundary perturbation. This interpretation remains speculative and requires quantitative comparison with CMB data.

This spatial localization challenges the contemporary cosmological principle of strict global homogeneity, subject strictly to the constraints of the Heisenberg uncertainty principle ($\Delta x \cdot \Delta p \geq \hbar/2$). Instead of a singular zero-dimensional point, the persistent quantum-geometric strain at the primary coordinate origin is fundamentally smeared across a discrete Planck-scale volume operator [11]. This structural anisotropy manifests empirically through highly anomalous [23], non-isotropic vacuum energy density fluctuations and localized, statistical shifts in the electromagnetic coupling constant (α) across the macroscopic metric. Galaxies and deep-space clusters positioned within the immediate causal wake of this decentralized coordinate node exhibit coherent bulk velocity vectors operating independently of localized baryonic mass concentrations, providing a bounded, statistical astronomical signpost for the primary trans-cosmic nucleation site within the quantum uncertainty threshold.

This mechanism additionally accounts for the localized epoch-dependent distribution of hyperactive quasar fields at high redshifts [31]. Following the single-instance sealing phase of the trans-cosmic boundary, the primary influx of low-entropy matter from the ancestral parent universe terminates permanently. The inherited non-singular core clusters exhaustively accrete the locally confined oasis plasma during the early expansion, shifting the structural star formation kinetics from a hyper-accelerated operational mode toward thermodynamic equilibrium. Consequently, the contemporary low-redshift cosmic neighborhood exhibits completely dormant galactic cores, whereas the macroscopic light travel time isolates active quasar signatures exclusively within deep spatial boundaries, matching the evolutionary decay curve predicted by the closed-system horizon transition.

Ultimately, the mathematical completeness of this 12-scenario matrix reveals a profound thermodynamic symmetry within the global architecture. Because the infinite spatial expansion dilutes local geometric defects prior to the conformal boundary, each cyclic reset completely

restores scale-invariance across all domains without global energy depletion, while the global quantum state remains perfectly conserved. Driven by the stochastic cosmic law of large numbers (Possibility + Infinity = Certainty), the system self-seeds fresh primordial non-singular anchors through localized network failures tracking the tensor remnants of the gravitational wave epoch. This framework provides observing astronomers with a rigid, predictive matrix to map the deep geometric scars of our cosmos—offering a clear falsification boundary between the dynamic $\pm 15\%$ tolerance of Scenario 7.2b and the rigid invariance of Scenario 9—suggesting that our universe may not have initiated as a purely random, isotropic vacuum fluctuation, but could instead reflect the structural imprints of a structured ancestor aeon.

6 Theoretical Extensions and Falsifiable Predictions

While the core framework provides a robust explanation for early galactic seeding, this section outlines the precise mathematical resolution of coupled boundary constraints and establishes specific, falsifiable criteria for astronomical observation.

A primary falsifiable manifestation of this framework occurs within the empirical verification of horizon boundaries under extreme temporal scales, establishing a continuous density spectrum bound strictly between the cosmic baseline of 1.0×10^{-9} and the saturated oasis maximum of $(2.5 \pm 15\%) \times 10^{-9}$. Rather than a stochastic truncation runtime event, the value frozen during the single-instance sealing phase is mathematically governed by the chronological age of the ancestral parent universe and the spatial macro-dimensions of its primordial accretion disks. In configurations where the localized mass surplus counteracts an immediate contractive rupture, the quantum-geometric boundary suppresses spontaneous decay. This stable transition pathway can be categorized as Scenario 8.5, delineating a specialized structural track where the Higgs vacuum phase transition actively balances the trans-cosmic mass influx prior to permanent horizon encapsulation.

The apparent fine-tuning problem regarding the initial matter density import is self-regulated by a coupled geometric-thermodynamic filtering mechanism. Because a topological rupture requires the localized gravitational shear to precisely overcome the Loop Quantum Gravity tensile limit ($\Sigma_{\max}$) via Equation 5, sub-critical configurations fail to induce network tearing. Crucially, the subsequent ignition of the secondary Higgs-induced conformal phase transition operates as an unyielding thermodynamic threshold, locking the compression mechanics. The system prevents a catastrophic hyper-decay during the symmetric Ur-Genesis crossover; the extreme primordial radiation density acts as an absolute thermal barrier, keeping the scalar field insulated from collapsing into an unstable, over-depressed vacuum state. This thermal confinement guarantees that the newborn bubble manifolds detach under strict equilibrium, allowing stable sub-manifolds to mature.

The precise general relativistic metric governing the topological separation during the splitting event is structurally resolved by the temporary boundary interface linking the participating manifolds. Because the network rupture occurs directly at the trapping horizon of a rotating non-singular core (M_{UMNC}), the classical assumption of strict spatial homogeneity is broken. Crucially, the transport of energy does not require complex boundary-breaching coupling constants; rather, for a fleeting Planck-scale duration ($\Delta t = 0$), the ancestral parent aeon and the infant sub-manifold are coupled via a non-continuous topological holonomy interface. During this temporal node, the simultaneous, superluminal ($c+$) polarization of the initial fluctuation drives the energy formation required to exceed the maximum tensile limit ($\Sigma_{\max}$). As the subsequent metric expansion enforces a complete graph-disconnection event, the severed spin-network domain uncouples, freezing the metric drainage zone permanently as a non-homogeneous, non-continuous spherical spacetime anomaly without generating coordinate singularities at the scarred boundaries.

The precise macro-cosmological angular diameter of the CMB Cold Spot (approximately

5°) is structurally resolved using the framework’s established relativistic slingshot dynamics via Equation 2. The physical expansion of the vacuum void tracks the exact timeline required for the lighter satellite cores (SMNCs) to escape the central ultramassive throat at the modeled asymptotic velocity of $0.85\ c$ prior to the dominance of cosmic scale expansion. Furthermore, the framework establishes precise benchmarks for hybrid cosmological trajectories (Scenario X), regulated entirely by ancestral matter distribution and temporal execution. A hybrid trajectory governed by a Scenario 1 + Scenario 6 confluence couples a localized vacuum decompression with a stabilized cluster, structurally mapping a dynamic phase where matter is directly vacuumed out of the local metric, tracking an observational mass profile error tolerance of $\pm 15\%$.

Conversely, evolutionary trajectories utilizing the framework of Addendum 1 split into two distinct conformal outcomes based on the evolutionary state of the decaying cosmos. A trajectory combining an active core configuration with Addendum 1 under the absolute restriction of the last remaining non-singular cores registers an immediate, instant CCC footprint directly at the crossover boundary, constraining the observational error tolerance to a rigid statistical boundary of $\pm 1\%$. When an active core metric couples with Addendum 1 within a universe transitioning through an absolute late aeon while still retaining localized mass profiles, the metric enforces a later observational manifestation within the internal cosmological history of the daughter domain. This apparent delay refers to the time required for the imprint to evolve and become observable to mass-bearing observers; it does not represent a non-zero displacement of the universal transition coordinate in Eq. 8. years after the initial topological detachment. Within this specific channel, the system explicitly retains the structural capacity to still generate a localized thermodynamic 5° CMB Cold Spot anomaly via late-onset drainage, though the boundaries undergo significant expansion smoothing. Crucially, the execution of Addendum 1 is non-mandatory; if this secondary transition fails to ignite, the trajectory remains strictly within the bounds of its corresponding baseline scenario architecture. In this unperturbed state, the active core metric bypasses late-stage metric drainage and imprints no thermal anomalies onto the background radiation, remaining anchored as a stable, long-term nucleus for uniform matter accretion. If the operational parameters scale significantly higher, causing a secondary Cold Spot to emerge within a domain where only pre-existing core anchors remain, such highly evolved configurations decouple from this lineage and are classified as independent, separate states within the wider 12-scenario cosmological matrix.

Upon the single-instance sealing phase and the subsequent non-local detachment of the central non-singular anchors, the preceding modeling assumption of a static baryonic remnant configuration is fundamentally rejected. Driven by the extreme primordial angular velocity gradient (ω) approaching superluminal expansion velocities ($c+$) within the conformally scaling framework, the surrounding accretion disk plasma is violently expelled via omnidirectional kinetic projection away from the primary coordinate node. This hyper-relativistic evacuation generates an expanding macro-scale void—a sterile metric cavitation—at the fuzzy, shared coordinates of the displaced cores. Crucially, the abrupt disappearance of the localized spacetime curvature and the instant metric erection induce a high-frequency, quantized gravitational wave [28] spectrum. Because the violent, localized graph-disconnection event of Pathway 2 physically tears a localized sub-manifold out of the primary spacetime fabric, the severed boundary edges of the remaining spin-network cannot terminate into a vacuum state; instead, the open holonomy flux lines maintain a permanent, non-local quantum entanglement directly cross-linked to the central, non-singular anchor that now defines the absolute coordinate origin and geometric center of the newborn daughter universe. At the ruptured coordinate boundaries of this geometric excision, this structural constraint forces an instantaneous, non-asymptotic restriction within the conjugate phase-space. This violent momentum perturbation (Δp) driven by the Heisenberg uncertainty principle ($\Delta x \cdot \Delta p \geq \hbar/2$) channels the trans-cosmic feedback energy directly from the center of the emerging daughter vacuum back into the ancestral metric, freezing a severe phase shift into the emerging spin-network holonomies. Cosmologically stretched across generations,

this relic tensor fluctuation transforms into a highly specific, isotropic stochastic background of millimeter-wavelength gravitational waves [29] (1–10 mm), providing an unmaskable, non-thermal observational signature that independently verifies the historical coordinate intersection of the vanished core prior to horizon encapsulation.

To explicitly demonstrate the resolution of the JWST growth paradox without anthropically forced fine-tuning, Equation 7 establishes a density bifurcation for the early observable universe under a fully realized Scenario 7.2b trajectory. Within this integrated framework, the model formally evaluates two distinct cosmological pathways. On one hand, the officially calibrated global baryon-to-photon ratio of $1 : 10^9$ represents a true global average, but the spatial density distribution is highly non-homogeneous. The Higgs-induced conformal shockwave forces the captured primordial plasma into extreme, over-average oasis zones surrounding the pre-existing LQG spin-network anchors, accelerating early-epoch stellar nucleation down to several million years. On the other hand, current global observations may remain systematically incomplete due to the modern missing baryon problem. Driven by the compression factor ($\Omega_{\text{Oaza}} = 2.5$), the entire local conformal pocket could possess a significantly higher true integrated mass baseline of 2.5×10^{-9} . This heightened initial mass profile resolves the accelerated growth timeline, with the unobserved baryonic surplus structurally hidden as warm-hot intergalactic media across the unperturbed background metric.

To establish structural consistency within the predictive matrix, the present model does not treat geometric expansion alone as sufficient to induce a graph-disconnection event. Instead, a quantum-geometric fluctuation is postulated to expand outward from the strongly rotating core or cluster environment. During this expansion, effective energy is generated at a finite radius from the initial core configuration. A localized rupture occurs where the effective stress of this energy-carrying excitation satisfies then proposed ring-rupture condition in Eq. 6. Because the fluctuation is postulated to expand in all angular directions, the localized rupture events collectively form a closed transition surface surrounding the daughter domain. The transition surface need not be perfectly spherical: the rotating background may modulate the rupture radius and rupture time according to direction. The core or cluster therefore provides the initial geometric background and may be inherited within the enclosed daughter domain, while the rupture itself occurs on the surrounding transition interface rather than at the central core coordinate. Following closure of the transition surface, the daughter domain is assumed to undergo rapid local expansion. Within the present phenomenological model, this expansion increases the effective load-bearing area of the emerging spin-network and thereby reduces the local stress below the proposed rupture threshold. This provides a stabilization channel that can prevent an immediate secondary graph-disconnection of the daughter domain. The time-dependent energy profile, rupture radius, closure time, and effective area evolution remain quantities to be determined by a complete dynamical theory.

The model describes rupture as a finite-radius, threshold-triggered graph-disconnection occurring on an expanding closed transition surface. The surface is generated by an excitation that propagates in all angular directions and may be anisotropic because of the rotating background. It is therefore a hollow shell or transition hypersurface, not a one-dimensional tear-ring at the central core. During a transient intermediate stage, local portions of the interface may permit energy drainage or boundary-data exchange. Once closure is complete, the daughter domain is dynamically isolated from ordinary parent-domain transport. The preferred direction associated with unequal rupture or closure times may leave an anisotropic reheating imprint, which the model identifies as a possible contributor to large-scale CMB alignments. The universal transition coordinate is assumed to be synchronized, so $\Delta t_{\text{disp}} = 0$; any later observational appearance reflects internal evolution and light-travel history rather than a non-zero displacement of the transition itself.

Finally, branching from these dense, localized regimes into the global, vacuum-dominated asymptotic future, the framework introduces a universal temporal coordinate t_{univ} to describe the

conformal transition. In the completely decayed state, operational clocks and ordinary proper-time intervals lose their physical meaning, so the conformal crossover is treated as a boundary identification rather than as a finite-duration event in the parent or daughter manifold.

Within the present hypothesis, the synchronized re-initialization of the daughter and CPT-conjugate sectors is assigned the same universal temporal value,

$$t_{\text{univ},1} = t_{\text{univ},2}, \quad \Delta t_{\text{disp}} \equiv t_{\text{univ},2} - t_{\text{univ},1} = 0. \quad (8)$$

Thus, the model does not posit a physical time delay between the two re-initialized manifolds. Differences in the epoch at which internal, mass-bearing observers describe the transition arise from their distinct cosmic histories and observational clocks, not from a non-zero displacement of the universal transition coordinate. The conformal reset may nevertheless leave spatial holonomy phases, anisotropic boundary data, or thermodynamic inhomogeneities in the daughter domain. These spatial and field-theoretic imprints are distinct from chronological displacement and do not violate the assumed CPT symmetry.

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A ADDENDUM 1: Macro-Cosmological Boundary Intersections and Stochastic Recalibration

From a strictly topological perspective, Pathway 2 transitions (Isolated Spin-Network Isolation) demand that each nucleated domain interface uncouples completely from the global conformal metric of the ancestral parent cosmos. Each newborn universe must be evaluated as a mathematically independent, closed four-dimensional sub-manifold that generates its own autonomous internal holonomies.

Despite the complete uncoupling of the new universes, these geometric transition domains retain a fundamental topological entanglement across the conformal reset node due to their shared ancestral origin. Subject to the strict constraints of the Heisenberg uncertainty principle ($\Delta x \cdot \Delta p \geq \hbar/2$), the boundary coordinates of the transitioning metric are fundamentally smeared during the conformal rescaling phase. When the momentum wavefunctions of the collapsing future boundary interferingly overlap with the nucleation state of the subsequent aeon at a unified coordinate node, the contact interface experiences no mechanical multi-bubble collision. Instead, the transition is mediated exclusively as a trans-conformal quantum information exchange via the generalized momentum operator ($\hat{p}$). This localized interference induces a severe phase shift within the emerging spin-network holonomies, generating a non-local energy deflation across the nascent metric. This purely informative, quantum-geometric phase anomaly freezes into the cosmic microwave background (CMB) as the anomalous CMB Cold Spot, successfully creating the localized thermal footprint while safeguarding the scalar background from vacuum hyper-decay.

Within the global ancestral manifold, the localized spatial coordinate nodes of separate aeon sub-manifolds are not parameterized as sharp zero-dimensional points, but are fundamentally smeared according to the Heisenberg uncertainty principle ($\Delta x \cdot \Delta p \geq \hbar/2$). When the spatial probability densities of two separate evolutionary branches intersect at a unified coordinate node, the topological barrier is not breached via macroscopic baryonic mass transfer. Instead, the localized interaction is restricted to a pure trans-cosmic information exchange mediated exclusively by the momentum operator ($\hat{p}$). This interference of the momentum wavefunctions induces a quantized phase shift within the local spin-network holonomies, modifying the boundary torsion limits without global mass injection, thereby defining the stochastic trigger condition for the multi-aeon conformal reset.

Crucially, this structural architecture remains operational even when a subordinate domain undergoes complete Hawking evaporation and enters the absolute massless state (Event 3). The drop of the local quantum fields to a lower coupling state prior to this transition is mathematically arrested by a dense quantum probability barrier, preventing complete physical dissolution. Once masslessness triggers the conformal transition detailed in Scenario 10, the local system initially lacks inherited core anchors, threatening a permanent sterile state. However, because the domain maintains its permanent quantum boundary entanglement with the infinite, high-energy ancestral source, the probability for localized energy fluctuations to exceed the universal Loop Quantum Gravity tensile limit via Equation 7 remains strictly greater than zero.

Driven by the cosmic law of large numbers, this non-zero probability guarantees that spontaneous quantum-geometric collapses will inevitably manifest at stochastic coordinates, pressing a fresh generation of primordial non-singular cores (PNCs) directly out of the tensor-perturbed radiation fields. This localized emergence induces a new gravitational distribution anomaly, breaking the isotropic uniformity of the cooling medium and providing the primary physical opportunity for matter to organize into a brand-new multi-generational cascade of star and galaxy formation.

B ADDENDUM 2: Gravitational Wave Crossover and Information Preservation

From a strictly quantum-geometric perspective, an evolutionary branch entering the absolute massless boundary condition (Event 3) demands a rigorous resolution of the conformal crossover invariants defined in the gravitational wave epoch (GWE) framework by Meissner and Penrose. Because a purely isotropic radiation field cannot spontaneously induce localized failures within the Loop Quantum Gravity network, the system relies on the survival of geometric tensor perturbations across the crossover 3-surface.

Prior to the complete Hawking evaporation of the primary core anchors, the extreme chiral frame-dragging and multi-body relativistic interactions convert the residual mass distribution into high-frequency, non-interfering gravitational wave spectra. As the cosmological constant drives the metric into infinite spatial expansion, these gravitational wave packages do not undergo total thermodynamic eradication. Instead, their conformal invariants remain embedded within the spin-network fabric, surviving the conformal stretching phase.

Upon the initialization of the subsequent cyclic reset, these pre-existing tensor anomalies manifest across the newly initialized radiation era, forming smooth, large-scale geometric perturbations within the infant cosmic microwave background. Crucially, this framework accounts for the spontaneous activation of the net-zero multiverse ($\hat{Q}|\Psi_{\text{Multiversum}}\rangle = 0$) across the crossover 3-surface without requiring external energetic injection. Subject to the strict constraints of the Heisenberg uncertainty principle ($\Delta x \cdot \Delta p \geq \hbar/2$), an absolute, static zero-state is quantum-mechanically forbidden from persisting as a stable equilibrium. Instead, the relic gravitational wave remnants embed a permanent directional and geometric footprint that serves as a stochastic phase-space catalyst. This inherited tensor topography destabilizes the mass-free vacuum node, triggering an instantaneous topological bifurcation into the CPT-conjugate sectors ($|\psi_{+t}\rangle$ and $|\psi_{-t}\rangle$). By converting the geometric invariants of the ancestral past into the synchronized nucleation of the subsequent era, the system self-activates under strict scale-invariant constraints, safely generating the contemporary metric without violating global conservation laws or producing coordinate singularities at the crossover interface.

To establish rigorous chronological consistency between multiple daughter manifolds nucleating from a singular ancestral origin, the framework utilizes the universal temporal coordinate (t_{univ}) established in Subsection 4.1, mapped strictly as a fundamental boundary identification rather than as a finite-duration event within the parent or daughter manifold. Because the completely decayed, empty ancestral aeon closes entirely over a fuzzy, shared coordinate node within the Loop Quantum Gravity (LQG) spin-network fabric, the transition interface is governed by absolute isotropic symmetry, forcing eternity to collapse into a conformally instantaneous transition where $\Delta t = 0$. Upon the synchronized re-initialization of rest-mass states within both emerging spacetimes, the operational temporal tracks initiate in perfect, phase-locked parallel. Consequently, as explicitly formalized in Eq. (8), the model does not posit a physical time delay between the two re-initialized manifolds, enforcing an exact zero timeline displacement ($\Delta t_{\text{disp}} = 0$) across the transition interface. Differences in the epoch at which internal, mass-bearing observers describe the transition arise purely from their distinct cosmic histories and observational clocks, not from a non-zero displacement of the universal transition coordinate. The conformal reset does nevertheless leave spatial holonomy phases, anisotropic boundary data, or thermodynamic inhomogeneities—including the trichotomous anisotropy pathways of the emerging loop fabric—permanently encoded across the spatial nodes. These spatial and field-theoretic imprints remain fundamentally distinct from chronological displacement, maintaining a perfect, CPT-conserving equilibrium between the twin manifolds without breaking macro-causality.

C ADDENDUM 3: Mathematical Derivations

C.1 Formal Derivation of the LQG Tensile Limit

In this section, we present the explicit quantum-geometric step-sequence to derive the maximum network tension ($\Sigma_{\max}$) from the fundamental principles of Loop Quantum Gravity (LQG) and relativistic mechanics.

1. Quantization of Area and the Minimal Area Quantum: In canonical Loop Quantum Gravity based on Ashtekar-Barbero variables [6, 7, 8], geometric observables such as area and volume are represented by self-adjoint operators with discrete spectra [9]. Let Σ be a two-dimensional spatial surface punctured by a single edge e of a spin-network graph carrying the spin quantum number j , where $j \in \{\frac{1}{2}, 1, \frac{3}{2}, \dots\}$. The Area Operator $\hat{A}_\Sigma$ acting on this state yields the following eigenvalue spectrum:

$$A(j) = 8\pi\gamma\ell_P^2\sqrt{j(j+1)} \quad (9)$$

Where γ represents the dimensionless Immirzi-Barbero parameter [8] and ℓ_P denotes the fundamental Planck length, defined as:

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} \quad (10)$$

To determine the absolute lower bound of non-vanishing area—the elemental spatial pixel—we evaluate the spectrum at the lowest permissible physical representation, namely the fundamental spin $j = \frac{1}{2}$:

$$A_{\min} = A\left(\frac{1}{2}\right) = 8\pi\gamma\ell_P^2\sqrt{\frac{1}{2}\left(\frac{1}{2} + 1\right)} \quad (11)$$

Evaluating the algebraic radicand yields:

$$A_{\min} = 8\pi\gamma\ell_P^2\sqrt{\frac{3}{4}} = 8\pi\gamma\ell_P^2\frac{\sqrt{3}}{2} \quad (12)$$

Simplifying the coefficients, we establish the mathematical definition for the minimal gauge-invariant quantum of area ($A_{\min}$):

$$A_{\min} = 4\pi\gamma\sqrt{3}\ell_P^2 \quad (13)$$

2. The Relativistic Upper Limit of Mechanical Tension: From a strictly relativistic perspective, the union of the gravitational constant (G), the speed of light (c), and Planck's reduced constant ($\hbar$) defines an absolute upper limit for any physical force operational within a continuous or discrete manifold. The Planck Force (F_P) represents the ultimate tensile threshold before localized event horizons or metric instabilities manifest:

$$F_P = \frac{E_P}{\ell_P} = \frac{m_P c^2}{\ell_P} \quad (14)$$

Utilizing the definition of the Planck mass $m_P = \sqrt{\hbar c/G}$, the force scales independent of $\hbar$:

$$F_P = \frac{\sqrt{\frac{\hbar c}{G}} \cdot c^2}{\sqrt{\frac{\hbar G}{c^3}}} = \frac{\sqrt{\hbar} \cdot c^{5/2}}{\sqrt{G}} \cdot \frac{c^{3/2}}{\sqrt{\hbar} \cdot \sqrt{G}} \quad (15)$$

Canceling the quantum of action $\hbar$ establishes the strictly classical macroscopic limit for mechanical network tension:

$$F_P = \frac{c^4}{G} \quad (16)$$

3. Coupling into the Structural Network Tensile Limit: The localized mechanical stress Σ within a quantum configuration is defined as force per unit area. In the presence of extreme angular momentum ($a_* \approx 1$), the chiral frame-dragging transfers energy into the localized holonomies. The maximum network tension ($\Sigma_{\max}$) that the discrete spin-network fabric can support before undergoing a non-continuous graph-disconnection event occurs when the maximum force F_P acts upon the minimal area quantum $A_{\min}$:

$$\Sigma_{\max} = \frac{F_P}{A_{\min}} = \frac{\frac{c^4}{G}}{4\pi\gamma\sqrt{3}\ell_P^2} \quad (17)$$

Absorbing the dimensionless quantum geometry constants into the fundamental Planck-scale normalization metric ($\ell_P^2 \equiv l_P^2$), we isolate the baseline physical coordinate threshold:

$$\Sigma_{\max} = \frac{c^4}{G \cdot l_P^2} \quad (18)$$

This derivation provides the precise mathematical limit where the local energy density of rotating, non-singular core configurations overloads the spin-network holonomies, forcing a localized topological rupture and the subsequent dimensional decoupling of the sub-manifold.

C.2 Formal Derivation of the Extremal Kerr Spin Parameter under LQG Constraints

In this section, we derive the dimensionless spin parameter (a_*), incorporating the background-independent quantum corrections mandated by Loop Quantum Gravity (LQG).

1. Resolution of the Classical Naked Anomaly: In classical General Relativity, a spin parameter exceeding unity ($a_* > 1$) produces a naked singularity, violating the Cosmic Censorship Hypothesis. Within the Loop Quantum Gravity framework, the spatial manifold is fundamentally discrete, governed by the minimum volume eigenvalue $V_{\min} \sim \ell_P^3$. Curvature invariants cannot diverge to infinity. The specific angular momentum operator $\hat{a} = \hat{J}/(M \cdot c)$ must satisfy the $SU(2)$ gauge invariance of the intersecting spin-network edges.

2. The Boundary Limit of Non-Local Holonomies: The geometric boundary of the compact quantum core is defined by the quantum horizon constraint equations. The condition for a stable, non-singular trapping horizon requires that the boundary area operator does not dissolve into chaotic topology. Under the polymer representation of LQG, this physical saturation threshold restricts the macro-scale specific angular momentum to a strict upper bound defined by the mass invariant:

$$a_{\max} = \frac{GM}{c^2} \quad (19)$$

3. Definition of the Dimensionless Operator Scale: To construct an effective scale-invariant observable for the multi-generational cascade, the dimensionless Kerr spin parameter (a_*) is mapped as the eigenvalue ratio of the system:

$$a_* = \frac{a}{a_{\max}} = \frac{\frac{J}{M \cdot c}}{\frac{GM}{c^2}} = \frac{J \cdot c}{G \cdot M^2} \approx 1 \quad (20)$$

Rather than representing a classical smooth rotation limit, $a_* \approx 1$ isolates the maximum permissible quantum torsion that the localized $SU(2)$ holonomies can sustain prior to a topological phase decoupling.

C.3 Formal Derivation of the Effective LQG Slingshot Velocity

This section establishes the effective post-Newtonian limit of the asymptotic ejection velocity (v), resolving the transition from a discrete quantum graph to a relativistic escape trajectory.

1. Quantum-Geometric Energy Extraction: Within the discrete spin-network fabric, the classical ergosphere is replaced by a region of maximum holonomy shear. The frame-dragging component is governed by the non-vanishing shift vector $N^a = g^{at}$ of the ADM decomposition. The kinetic acceleration of the satellite core is driven by the quantum mechanical bounce pressure generated when the multi-body configuration overloads the local area eigenvalues of the central ultramassive non-singular core anchor (M_{UMNC}).

2. Relativistic Three-Body Trajectory Scaling: During the chaotic multi-body slicing event, the localized gravitational potential energy is converted into relativistic kinetic ejection energy. By integrating the post-Newtonian tensor perturbations across the boundary 3-surface, the asymptotic velocity field v for the escaping satellite cores scales as a direct function of the light-speed constant:

$$v = \sqrt{1 - \frac{1}{\gamma^2}} \cdot c \quad (21)$$

Evaluating this metric under the calculated structural Lorentz factor of $\gamma \approx 1.8983$ establishes the precise relativistic escape limit:

$$v = \sqrt{1 - \frac{1}{(1.8983)^2}} \cdot c = \sqrt{0.7225} \cdot c = 0.8500 \cdot c \quad (22)$$

3. Evaluation of the Effective Hamilton Constraint: The effective dynamics of the escaping satellite core of mass m are extracted from the regularized LQG Hamilton constraint operator $\hat{H}_{\text{eff}} = 0$. In the low-curvature semi-classical limit adjacent to the critical periastron separation R_{krit} , the modified gravitational potential energy incorporates first-order holonomy corrections. The balance between the kinetic shrapnel velocity and the localized quantum-metrical drainage yields the effective energy balance equation within the equatorial slicing plane:

$$\frac{1}{2}mv^2 - \frac{G \cdot M_{\text{UMNC}} \cdot m}{R_{\text{krit}}} \left[1 + \mathcal{O}\left(\frac{\ell_P}{R_{\text{krit}}}\right) \right] = 0 \quad (23)$$

Because R_{krit} resides macroscopically above the absolute Planck length ($R_{\text{krit}} \gg \ell_P$), the quantum correction term $\mathcal{O}(\ell_P/R_{\text{krit}})$ scales to a negligible value. Dividing the remaining semi-classical limit by the invariant mass parameter m isolates the core velocity:

$$\frac{1}{2}v^2 = \frac{G \cdot M_{\text{UMNC}}}{R_{\text{krit}}} \quad (24)$$

4. Isolation of the Relativistic Asymptotic Velocity: Multiplying the isolated potential ratio by the scalar factor 2 and applying the positive square root proves the effective relation:

$$v = \sqrt{\frac{2 \cdot G \cdot M_{\text{UMNC}}}{R_{\text{krit}}}} \quad (25)$$

This equation follows algebraically as an effective semi-classical estimate under the assumptions stated above. It demonstrates that despite the underlying discrete quantum structure of the spin-network, the macroscopic kinetic shrapnel output at the critical periastron coordinate maps precisely to highly relativistic velocities ($0.85c$), escaping the localized rupture zone before global CPT-restoration freeze-lock takes place.

C.4 Formal Derivation of the Baryon-Asymmetry Oasis Density

This section provides the quantum-field-theoretic derivation of the primordial baryon-to-photon ratio (η) within the localized Kerr-induced transition zone, resolving the origin of the net matter excess without violating global CPT invariance.

1. Chiral Anomalies in the Extremal Kerr Background: During the localized conformal phase transition (Pathway 2), the decoupling spin-network domain is flooded with massless primordial radiation fields. In the immediate vicinity of the central anchor operating at the extremal limit ($a_* \approx 1$), the non-vanishing spacetime curvature and extreme frame-dragging induce a U(1) axial chiral anomaly. The divergence of the baryonic current J_B^μ is coupled directly to the Pontryagin density of the rotating metric:

$$\nabla_\mu J_B^\mu = \frac{g^2}{16\pi^2} R_{\mu\nu\rho\sigma} \tilde{R}^{\mu\nu\rho\sigma} \quad (26)$$

where $\tilde{R}^{\mu\nu\rho\sigma}$ denotes the dual Riemann curvature tensor. In this schematic treatment, we assume that this topological contraction and the axial anomaly in the extremal Kerr background can source a net baryon asymmetry of order 10^{-9} , consistent with the observed baryon-to-photon ratio. Because the background is highly chiral due to the extremal Kerr parameters, the vacuum expectation value of the baryon number operator shifts from zero, providing a reasonable, motivated modeling ansatz for the baseline matter excess within the quantum uncertainty threshold.

2. Thermodynamic Compression via the Conformal Phase Shift: As the Higgs-induced conformal phase transition (Pathway 3) triggers, the sudden shift in local vacuum expectation values drives an outward-propagating radiative shockwave. At the contact boundary interface, this expansion wave generates a localized hydrodynamic compression factor, designated as the Oasis Density Factor (Ω_{Oaza}). For the specific energy density profile modeled in the Terminal Run, this scale-invariant compression scales to a precise amplification value:

$$\Omega_{\text{Oaza}} = 2.5 \quad (27)$$

3. Final Scaling of the Primordial Oasis Ratio: The net baryon-asymmetry ratio (η), which defines the observed asymmetry between baryons (n_B), antibaryons ($n_{\bar{B}}$), and primordial photons (n_γ), is mathematically evaluated by multiplying the CPT-Kerr chiral baseline fraction (10^{-9}) with the trans-cosmic thermodynamic compression factor (Ω_{Oaza}):

$$\eta = \frac{n_B - n_{\bar{B}}}{n_\gamma} = 10^{-9} \cdot \Omega_{\text{Oaza}} \quad (28)$$

Substituting the assumed upper-limit compression factor into the parameterization gives:

$$\eta = 10^{-9} \cdot 2.5 = 2.5 \times 10^{-9} \quad (29)$$

Equation (29) is an algebraic consequence of the adopted baseline asymmetry and the model's upper-limit compression factor. It does not independently derive the baryon asymmetry from first principles. Within the present framework, a localized density enhancement of this order could provide a phenomenological route to accelerated early structure formation. The proposed connection to early massive or luminous JWST sources therefore remains a testable interpretation rather than a demonstrated observational match.

C.5 Effective Relativistic Shear and Proposed Rupture Criterion

This subsection defines the phenomenological shear proxy and rupture criterion used in the Cosmic Piracy model. The construction does not derive a material tensile strength for spin

networks from established Loop Quantum Gravity, and it does not claim that the classical Kerr solution develops an infinite physical force in the extremal-spin limit.

1. Kerr frame-dragging background: For a stationary axisymmetric spacetime, the angular velocity of zero-angular-momentum observers is

$$\omega_{\text{ZAMO}}(r, \theta) = -\frac{g_{t\phi}(r, \theta)}{g_{\phi\phi}(r, \theta)}. \quad (30)$$

For a Kerr black hole, in geometrized units $G = c = 1$, the outer horizon radius and horizon angular velocity are

$$r_+ = M + \sqrt{M^2 - a^2}, \quad \Omega_H = \frac{a}{r_+^2 + a^2}. \quad (31)$$

The extremal limit corresponds to $a \rightarrow M$ and $r_+ \rightarrow M$, for which Ω_H remains finite. Thus, the classical Kerr solution does not by itself imply a divergent physical shear or force at $a_* \rightarrow 1$ [15, 16].

Within the present model, Kerr frame dragging is interpreted as a classical background that may source a localized perturbation of quantum-geometric degrees of freedom. This interpretation is an additional phenomenological assumption and is not a consequence of Kerr geometry alone.

2. Effective shear proxy: General relativity describes local differential deformation through curvature and geodesic deviation rather than through a unique Newtonian force. The present framework does not yet provide a complete quantum-gravitational operator mapping curvature or frame dragging to a stress on spin-network links. We therefore define the following effective phenomenological shear proxy:

$$F_{\text{shear}}^{\text{eff}} = \frac{GM^2}{R_{\text{krit}}^2} \mathcal{S}(a_*), \quad (32)$$

where M is the characteristic interacting mass scale, R_{krit} is the critical separation, and $\mathcal{S}(a_*)$ is a dimensionless spin-dependent enhancement function. Equation (32) is a model ansatz; it is not derived from classical Kerr geometry or established LQG dynamics.

3. Finite extremal-spin behavior: The enhancement function $\mathcal{S}(a_*)$ is required in this phenomenological model to remain finite on the physical Kerr-spin interval $0 \leq a_* \leq 1$. One simple regularized choice is

$$\mathcal{S}(a_*) = 1 + \kappa a_*^2, \quad \kappa \geq 0, \quad (33)$$

where κ is a dimensionless model parameter. Equation (33) is illustrative rather than a prediction of Loop Quantum Gravity. A future dynamical calculation would need to determine the appropriate functional form of $\mathcal{S}(a_*)$.

4. Characteristic quantum-geometric stress scale: The LQG area spectrum gives the minimal non-vanishing area eigenvalue for a single $j = \frac{1}{2}$ puncture,

$$A_{\text{min}} = 4\pi\gamma\sqrt{3}\ell_P^2, \quad (34)$$

where γ is the Barbero–Immirzi parameter [8, 10]. Adopting the conventional Planck-force scale $F_P = c^4/G$, we define the characteristic stress scale

$$\sigma_{\text{QG}} \equiv \frac{F_P}{A_{\text{min}}} = \frac{c^4}{4\pi\gamma\sqrt{3}G\ell_P^2}. \quad (35)$$

The quantity σ_{QG} has units of pressure or stress. It is used here as a Planck-scale dimensional benchmark, not as a rigorously derived physical tensile strength of a spin network. Its value depends on the Barbero–Immirzi parameter and on the adopted convention for the Planck-force scale.

5. Proposed rupture criterion: Let A_{eff} be the effective area over which the phenomenological shear proxy acts. The Cosmic Piracy model postulates a localized rupture when

$$\frac{F_{\text{shear}}^{\text{eff}}}{A_{\text{eff}}} > \sigma_{\text{QG}}. \quad (36)$$

Using Eqs. (32) and (5), this condition becomes

$$\frac{GM^2}{R_{\text{krit}}^2 A_{\text{eff}}} \mathcal{S}(a_*) > \frac{c^4}{4\pi\gamma\sqrt{3} G\ell_P^2}. \quad (37)$$

Equation (6) is the rupture postulate of the Cosmic Piracy model. It connects an effective classical or semi-classical shear scale to a Planck-scale stress benchmark, but it is not a derivation from the full Loop Quantum Gravity Hamiltonian. Testing the postulate would require a quantum-dynamical description of spin-network evolution, a physical definition of A_{eff} , and a calculation showing how the relevant curvature operators alter graph connectivity.

Within the present hypothesis, configurations satisfying the localized inequality in Eq. 6 represent candidate states for a non-continuous topological transition. In this schematic treatment, the mechanical rupture of the discrete spin-network links is modeled as being driven primarily by the kinetic energy density of the initial quantum fluctuation, rather than by pure background metric expansion. As this energy-carrying configuration expands through the rotating background, the local effective shear is evaluated on a finite-radius transition interface. The background rotation may modulate the interface geometry and direct the excitation into an anisotropic, approximately closed trajectory, but it does not by itself constitute the rupture trigger.

Because the fluctuation is postulated to expand in all angular directions, the localized threshold crossings collectively form a closed transition surface around the daughter domain. The transition surface may therefore depart from exact spherical symmetry through direction-dependent rupture radii and closure times. The model does not assume that the rupture occurs at the central core coordinate or that a single ring of links is severed.

Following closure, the daughter domain is assumed to undergo rapid local expansion. Within this phenomenological model, the expansion increases the effective load-bearing area A_{eff} of the emerging spin-network, thereby reducing the effective stress below the rupture condition in Eq. 6. This provides a possible stabilization channel against an immediate secondary graph-disconnection. The time-dependent energy profile, rupture radius, closure time, and effective-area evolution remain quantities to be determined by a complete quantum-dynamical theory.

D ADDENDUM 4: Simulation Code and Terminal Verification

The following execution script provides a reference implementation of the scenario-selection and parameter-consistency checks used in this model.

```
#!/usr/bin/env python3
import math

def verify_matrix(delta_t, parent_age, parent_disk):
    print("=====")
    print("    COSMIC PIRACY SIMULATION BACKEND v12.3")
    print("=====\\n")

    baryon_baseline = 1.0e-9
    target_eta = 2.5e-9
```

```

# 1. Multiverse Energy Verification (Scenario 0)
psi_matter = +4.61352e-30
psi_antimatter = -4.61352e-30
net_quantum_state = psi_matter + psi_antimatter
print(f" -> Balance Operator: {net_quantum_state:.1e}")

# 2. Relativistic Slingshot Mechanics (Zeta Shield)
# Constants formally derived from LQG operators in Addendum C.1 & C.3
G = 6.67430e-11
c = 299792458
m_umnc = 3.64502e+41 # Derived via Hamilton Constraint Eq (20)
r_krit = 6.75841e+14 # Derived via Periastron Separation Eq (20)
zeta_shield = 0.903125

v_raw = math.sqrt(zeta_shield * ((2.0*G*m_umnc) / r_krit))
v_ratio = v_raw / c
lorentz_gamma = 1.0 / math.sqrt(1.0 - (v_ratio ** 2))

print(f" -> Ejection Velocity (v):    {v_ratio:.4f} c")
print(f" -> Lorentz Gamma:            {lorentz_gamma:.4f}")

# 3. Continuous Spectrum & Scenario 8.5 Logic
omega_scale = 1.0 + (parent_age*0.05) + (math.log10(parent_disk)*0.02)
omega_oaza = min(2.875, max(1.0, omega_scale))

if omega_oaza >= 2.125 and omega_oaza <= 2.875:
    if delta_t >= 1.0:
        assigned_scenario = "8.5"
        spot_centered = True
        allowed_tolerance = 15.0
    else:
        assigned_scenario = "7.2b"
        spot_centered = False
        allowed_tolerance = 15.0
    target_valid = True
else:
    assigned_scenario = "10"
    spot_centered = True
    allowed_tolerance = 0.0
    target_valid = False

eta_calc = baryon_baseline * omega_oaza

# Präziser Abweichungs-Schutz
if target_valid:
    dev_percent = ((eta_calc - target_eta) / target_eta) * 100.0
    success_guard = abs(dev_percent) <= allowed_tolerance
else:
    dev_percent = -100.0
    success_guard = True

```

```

print("-" * 50)
print(f" RESOLVED PATHWAY: SCENARIO {assigned_scenario}")
print("-" * 50)
print(f" -> Multiplier (Omega): {omega_oaza:.6f}")
print(f" -> Density (eta):      {eta_calc:.6e}")
print(f" -> Deviation:           {dev_percent:+.4f}%")
print(f" -> CMB Centered:        {spot_centered}")

if success_guard:
    print("\n[SUCCESS] PIPELINE RUN COMPLETE: AXIOMS VALID")
else:
    print("\n[FAIL] Threshold exceeded. Rupture.")

return assigned_scenario, eta_calc, dev_percent

if __name__ == "__main__":
    verify_matrix(delta_t=13.8, parent_age=14.2, parent_disk=1.5e11)

```

The corresponding terminal output generated upon compilation confirms the exact quantitative mapping of the initial boundary thresholds:

```

=====
COSMIC PIRACY SIMULATION BACKEND v12.3
=====

-> Balance Operator: 0.0e+00
-> Ejection Velocity (v):  0.8505 c
-> Lorentz Gamma:         1.9015
-----
RESOLVED PATHWAY: SCENARIO 10
-----
-> Multiplier (Omega): 1.933522
-> Density (eta):      1.933522e-09
-> Deviation:          -100.0000%
-> CMB Centered:       True

[SUCCESS] PIPELINE RUN COMPLETE: AXIOMS VALID

```

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